Researchers discover simple fix to eliminate energy loss in fusion plasma heating

核心词汇

- **dissipate** v. 浪费；使... 消散
- **sum** v. 合计，总计 n. 总数，和；金额；算术题
- **helpful** adj. 有帮助的，有益的，有用的
- **finally** adv. 最后,最终
- **simulate** v. 模仿,模拟；假装,冒充
- **interference** n. 干扰，冲突；干涉
- **response** n. 响应；反应；回答
- **disorder** n. 失调,疾病；混乱,杂乱；骚乱
- **active** adj. 活跃的，敏捷的；积极的；在活动中的
- **select** v. 选择,挑选 adj. 精选的,选择的
- **consequently** adv. 结果，因此，所以
- **composition** n. 组成,成分；〔音乐、诗歌或文字〕作品；作文
- **application** n. 应用；申请；应用程序；敷用
- **entail** v. 使需要，必需；承担；遗传给；蕴含 n. 引起；需要；继承
- **unit** n. 单位，单元；装置；[军] 部队；部件
- **avail** n. 效用，利益 v. 有益于，有益于；使对某人有利。
- **observe** v. 庆祝
- **fade** v. 褪色；凋谢；逐渐消失 adj. 平淡的；乏味的 n. [电影][电视] 淡出；[电影][电视] 淡入
- **likelihood** n. 可能性，可能
- **successive** adj. 接连的,连续的
- **fitting** n. 装配，装置；试穿，试衣 adj. 适合的，适宜的；相称的
- **complete** adj. 完整的；完全的；彻底的 v. 完成
- **gather** v. 收集；收割；使... 聚集；使... 皱起 n. 聚集；衣褶；收获量

- **overcome** v. 克服；胜过
- **quantitative** adj. 定量的；量的，数量的
- **hydrogen** n. [化学] 氢
- **relate** vt. 叙述，讲述；使联系
- **motion** n. 运动,动；提议,动议 v. 提议,动议

双语正文

For decades, nuclear fusion researchers have chased a tiny but transformative tweak: a way to ensure every watt of energy dumped into a tokamak's superheated soup actually fuels reactions instead of going to waste. That goal may finally be within reach, thanks to a team led by the U.S. Department of Energy's Princeton Plasma Physics Laboratory (PPPL). In response to long-standing frustrations with dissipated heating power, the group has used advanced simulations to confirm a near-zero-cost, fitting solution that could shave years off fusion's commercialization timeline.

几十年来，核聚变研究者一直在追求一个微小却能带来变革的调整方向：一种能让注入托卡马克超高温等离子体"浓汤"的每瓦特能量都真正用于驱动聚变反应，而非白白浪费的途径。得益于美国能源部普林斯顿等离子体物理实验室（PPPL）领导的团队，这一目标终于有望实现。针对长期困扰研究者的耗散加热功率问题，该团队借助先进模拟技术，确认了一种近乎零成本的可行方案，有望将聚变商业化的时间线缩短数年。

The problem stems from unwanted, unhelpful wave interference generated when helicon waves—the primary tool for deep plasma heating—blast from an antenna toward a tokamak's doughnut-shaped, magnetically confined core. Unlike their helpful helicon cousins, these errant "slow modes" cannot weave through tight magnetic field lines to reach the hydrogen-rich heart where fusion reactions thrive. Instead, they linger near the antenna's edge, dampen or fade quickly, and consequently gobble up energy without any tangible avail. Previous 1D models hinted at a workaround, but it took quantitative 2D Petra-M code—used for both fusion and space plasma motion studies—to validate it and pinpoint the exact sweet spot.

问题源于螺旋波——用于深度等离子体加热的主要工具——从天线射向托卡马克甜甜圈形状的磁约束核心时产生的无用、有害的波干涉。与发挥有益作用的螺旋波不同，这些偏离正轨的"慢模"无法穿过紧密的磁场线，抵达聚变反应蓬勃发生的富氢核心区域。它们只会滞留在天线边缘附近，迅速衰减或消散，因此白白消耗能量，毫无实际用处。此前的一维模型曾暗示存在一种解决办法，但直到借助用于聚变和空间等离子体运动研究的定量二维 Petra-M
代码，研究者才验证了该方案并精准找到了最佳参数。

The Petra-M simulations, replicated DIII-D tokamak conditions (operated by General Atomics in San Diego), tested three variables: antenna alignment, the orientation of a protective metal unit called a Faraday screen, and the electron composition of the thin plasma layer just ahead of the antenna. After successive virtual trials, the team's data gathered that the Faraday screen's tilt was the critical lever. Unlike prior assumptions that required a complete redesign of antenna systems, the fix entailed only angling the screen five degrees or less relative to the antenna's slats. At that precise threshold, the screen effectively short-circuits slow-mode formation, making them fizzle out before they can propagate inward. If the tilt exceeds even half a degree past five, slow modes surge back—active and destructive—revealing a sensitivity that surprised even senior PPPL physicist Masayuki Ono, a co-author. "We thought we might see gradual changes, but this was like flipping a light switch," Ono noted. "One small nudge, and the energy loss overcome almost entirely."

Petra-M 模拟复制了由圣地亚哥通用原子公司运营的 DIII-D
托卡马克的实验条件，测试了三个变量：天线对准度、被称为法拉第屏的保护性金属装置的朝向，以及天线前方薄层等离子体的电子成分。经过连续的虚拟试验，团队收集的数据显示，法拉第屏的倾斜角度是关键调控因素。与此前认为需要彻底重新设计天线系统的假设不同，该解决方案仅需要将法拉第屏相对于天线板条倾斜五度或更小的角度。在这一精确阈值下，法拉第屏能有效阻断慢模的形成，使慢模在向内传播前就逐渐消失。哪怕倾斜角度比五度超出半度，慢模就会重新活跃并造成破坏——这种敏感性甚至让作为共同作者的 PPPL

资深物理学家小野正幸都感到惊讶。"我们原本以为只会看到渐进式的变化，但这次的情况就像按动电灯开关一样干脆利落，"小野说道，"只需要小小的调整，能量损失就几乎完全被消除了。"

Eun-Hwa Kim, lead author and PPPL principal researcher, emphasized that this is the first complete 2D exploration of slow-mode mitigation, adding that the results could soon translate to real-world application. "Instead of throwing more money at bigger antennas or more power sources, we can select the optimal screen angle for existing facilities first," she said. "That's a game-changer for projects racing to hit net energy gain." The team now plans to refine their models to account for more plasma properties—such as ion density and magnetic field disorder—and will eventually collaborate with General Atomics to test the five-degree tilt on DIII-D hardware. If those physical trials match the Petra-M sum of virtual data, the likelihood of commercial fusion reaching the grid by the 2040s will rise dramatically.

论文第一作者、PPPL 首席研究员金恩和（Eun-Hwa Kim）强调，这是首次对慢模抑制进行完整的二维探索，并补充说研究结果很快就能转化为实际应用。她说："与其投入更多资金打造更大的天线或更多的电源，我们可以先为现有设施选择最优的屏体倾斜角度。这对于竞相实现净能量增益的项目来说，是一个改变游戏规则的突破。"该团队目前计划优化模型，纳入更多等离子体特性参数——比如离子密度和磁场紊乱程度——并最终将与通用原子公司合作，在 DIII-D 装置上测试五度倾斜方案。如果这些实体试验与 Petra-M 的虚拟数据总和相符，那么商业化聚变在 21 世纪 40 年代接入电网的可能性将大幅提升。

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